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Hybrid Deep Eutectic Solvents with Flexible Hydrogen-Bonded Supramolecular Networks for Highly Efficient Uptake of NH_3

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Serious environmental concerns have led to a great demand for efficient uptake of NH_3 by solvents. However, traditional aqueous absorbents have many shortcomings and efforts to use ionic liquids have met with limited success. A hybrid deep eutectic solvents (DESs) designed with a flexible hydrogen-bonded supramolecular network exhibits both exceptional NH_3 uptake capacity and superior desorption–regeneration performance, along with superb NH_3/CO_2 selectivity and environmental merit. Elucidated by molecular dynamic simulations and spectroscopic analysis, the abundant hydrogen-bonding sites in the hybrid DESs bind every atom of the NH_3 molecule and enable strong physical reversible solvation, whereas the multiple interactions among the hybrid components create a flexible hydrogen-bonded supramolecular network and allow for solvent-unbreaking absorption to ensure the full participation of the solvent and process stability. A mass solubility of NH_3 up to 0.13 $g\,g^{-1}$ was achieved at 313 K and 101 kPa by the hybrid DES choline chloride/resorcinol/glycerol (1:3:5), which is higher than all reported ionic liquids and ordinary DESs. Moreover, the performance remained the same after ten absorption–desorption cycles and the DESs could be easily regenerated.

Introduction

Global challenges in environmental and sustainable chemistry have created a great demand for efficient uptake of NH_3 from the gas phase, because NH_3 is not only a major alkaline gaseous atmospheric pollutant and an important precursor of PM2.5 (particulate matter $\leq$ 2.5 μm), but also the dominant working medium in low-hazard Freon-free absorption refrigerators.[1] The omnipresent absorbents in current NH_3 capture plants and refrigerators are water and aqueous inorganic or carboxylic acids, which have suffered from lots of longstanding problems.[2] First, the high volatility of water and many acids and the strong reactivity of NH_3 with those solvents incurs significant difficulties to the desorption of NH_3 and recycle of absorbents, making the energy-intensive distillation and a complex process usually indispensable in the plants and refrigerators.[3] Moreover, strong corrosion of equipment by the absorbents often occurs, and the wastewater discharge is hard to avoid. A lot of efforts have been exerted to address these problems, especially the utilization of ionic liquids as low-volatile solvents in absorption.[4] However, limited success has been achieved owing to the unsatisfactory mass solubility of NH_3 and the complex synthesis of relatively expensive task-specific ionic liquids.[5] Thus far, it remains a big challenge to develop new materials for NH_3 absorption to achieve both high NH_3 solubility and superior recyclability along with environmental and synthetic merits.

In recent years, deep eutectic solvents (DESs) have been put forward as a new class of solvents[6] that share the advantages of ultralow volatility and designability of ionic liquids and are superior in price, synthesis, and biodegradability.[6e] However, one chemical characteristic of DESs that is distinct from common solvents and ionic liquids is that they are supramolecular complexes of different individual components rather than a "pure" substance.[7] Therefore, DESs are more "fragile" liquids and their structure may decompose into individual components if there is a strong interaction with other solvents or solutes. Unfortunately, this fragile nature of DESs has not been paid enough attention up to now, although it is essential for their practical performance. An attempt to increase the efficiency of a DES for dissolving a solute by enhancing the DES–solute interactions, which is the common-sense strategy for ordinary solvents and ionic liquids, will probably fail because the enhanced DES–solute interactions may also break up the supramolecular structure of the DES and generate nonfunctional solids. This trade-off between the DES–solute interactions and the DES stability is particularly significant if the solute molecules are very active, as is the case for NH_3 (Figure 1).

Herein, we report for the first time the design of novel hybrid DESs with a flexible hydrogen-bonded supramolecular network that enables both excellent NH_3 uptake capacity and

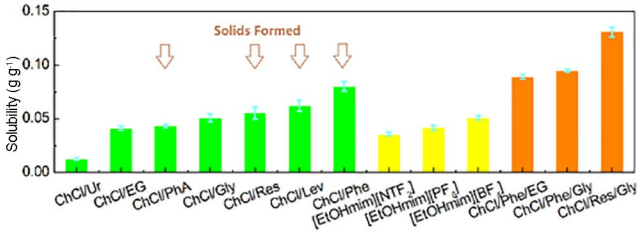
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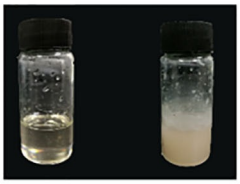
a



b

Composition		Molar Ratio
ChCl/Ur	Choline Chloride/Urea	1:2
ChCl/EG	Choline Chloride/Ethylene Glycol	1:2
ChCl/PhA	Choline Chloride/Phenylacetic Acid	1:2
ChCl/Gly	Choline Chloride/Glycerol	1:2
ChCl/Res	Choline Chloride/Resorcinol	1:3
ChCl/Lev	Choline Chloride/Levulinic Acid	1:2
ChCl/Phe	Choline Chloride/Phenol	1:3
ChCl/Phe/EG	Choline Chloride/Phenol/Ethylene Glycol	1:3:5
ChCl/Phe/Gly	Choline Chloride/Phenol/Glycerol	1:3:5
ChCl/Res/Gly	Choline Chloride/Resorcinol/Glycerol	1:3:5

c



d

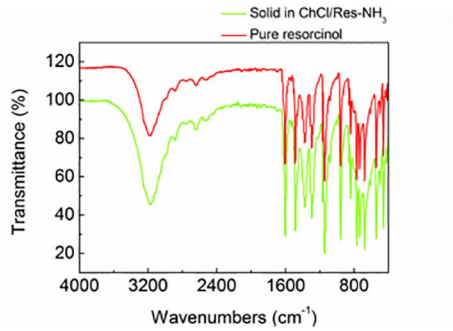


Figure 1. a) NH_3 solubilities in various solvents. The DESs that form solids are marked by arrows. b) The composition and molar ratio of the DESs in Figure 1a. c) Images of the fresh ChCl/Res (left) and ChCl/Res after making contact with NH_3 (right). d) FTIR of the white solid formed in the ChCl/Res–NH_3 system and pure resorcinol.

superior desorption–regeneration performance to set a new benchmark for reversible NH_3 absorption by solvents. The abundant hydrogen-bonding sites in the hybrid DESs grasp every atom of the NH_3 molecule and enable a strong physical reversible solvation, whereas multiple interactions among the hybrid components create a flexible hydrogen-bonded supramolecular network and allow for solvent-unbreaking absorption to ensure the full participation of the solvent and process stability. Consequently, the trade-off between the DES–solute interactions and the DES stability is removed. An exceptional NH_3 mass solubility of 0.13 g g^{-1} at 313 K and 101 kPa was observed by the hybrid DES composed of choline chloride/resorcinol/glycerol (1:3:5; ChCl/Res/Gly), which was considerably higher than that of all reported ionic liquids and ordinary DESs.

Results and Discussion

NH_3 absorption

Several widely used ordinary DESs, which were binary mixtures of choline chloride (ChCl) and different hydrogen bond donors (HBDs), were first tested for the absorption of NH_3 at 313 K and 101 kPa (Figure 1). White solids quickly appeared during the absorption by DESs with acidic HBDs (phenols and carboxylic acids), which severely inhibited the contact of NH_3 with the remaining absorbent and prevented the continuation of the absorption process. FTIR measurement of these solids revealed that they were almost pure phenols (Figure 1 d, exemplified by the absorption of 1:3 ChCl/resorcinol), indicating that the solid formation is caused by the breaking up of the supramolecular structure in the DES. The adverse effect of this phenomenon is well demonstrated by the abnormal acidity–capacity relationship in the absorption by those solvents. Given that phenols and carboxylic acids are much more acidic than alcohols and urea, the NH_3 uptake capacities of DESs containing phenols and carboxylic acids were expected to be higher than those of ChCl/Gly, ChCl/EG, and ChCl/Ur. However, these expected results were not observed. The NH_3 uptake capacity of ChCl/PhA was lower than that of ChCl/Gly. Moreover, although resorcinol contains more hydroxy groups than phenol and is also more acidic (the respective pKa values of resorcinol and phenol are 9.44 and 9.94), the NH_3 uptake capacity of ChCl/Res (0.053 g g^{-1}) was unexpectedly much lower than that of ChCl/Phe (0.081 g g^{-1}) owing to the more extensive formation of solids. This abnormal acidity–capacity relationship indicated that, because of the fragile supramolecular structure of DESs, the well-acknowledged strategy to enhance the solubility of basic solutes by increasing the acidity or the number of acidic groups of the solvents is not necessarily valid in the case of DESs. The use of neutral HBDs instead of phenols and carboxylic acids can prevent this structural breakage to ensure a "complete" absorption, but the reduced solute–solvent interactions also reduces the NH_3 uptake capacity.

To address the trade-off between the NH_3–DES interaction and the supramolecular stability, we prepared a novel class of hybrid DESs, which was formed from a ternary mixture of choline chloride, phenolic compounds, and polyalcohol and can be deemed as a hybrid of two binary DESs (choline chloride + phenolic compound and choline chloride + polyalcohol). Compared with ordinary DESs, the hybrid DESs reached a high level of NH_3 solubility without the precipitation of solids. The uptake capacities of NH_3 in choline chloride/phenol/ethylene glycol (1:3:5, molar ratio, ChCl/Phe/EG) and choline chloride/phenol/glycerol (1:3:5, molar ratio, ChCl/Phe/Gly) at 313 K and 101 kPa were 0.091 g g^{-1} and 0.095 g g^{-1}, respectively, which were superior to the related binary analogues, ChCl/Phe (0.082 g g^{-1}) and ChCl/Gly (0.053 g g^{-1}). More notably, when phenol in ChCl/Phe/Gly was replaced by resorcinol, the NH_3 solubility in ChCl/Res/Gly (1:3:5 molar ratio) improved further

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and reached up to 0.12 g g^{-1}, which was significantly higher than the ordinary DESs. Among the various investigated ionic liquids in earlier reports of NH_3 absorption,[4b,g] 1-hydroxyethylimidazolium tetrafluoroborate ([EtOHmim][BF_4]) showed the highest NH_3 mass solubility. However, the value at 313 K and 101 kPa was 0.051 g g^{-1} (Figure 1a), which was only 39% of that in ChCl/Res/Gly. The other top-performing ionic liquids had even lower mass solubilities, 0.036 g g^{-1} for 1-hydroxyethylimidazolium bis(trifluoromethane)sulfonimide ([EtOHmim][NTf_2]) and 0.042 g g^{-1} for 1-hydroxyethylimidazolium hexafluorophosphate ([EtOHmim][PF_6]).

Table 1 lists all of the ionic liquids that have been reported for NH_3 absorption and the units of the solubilities were unified in g(NH_3) g(solvent) $^{-1}$. The NH_3 solubilities in all those ionic liquids were lower than that in ChCl/Res/Gly under the same conditions. By comparing the NH_3 solubilities among different ionic liquids, the incorporation of hydroxy or an acidic $N^{\delta+}$–H group in ionic liquids was found to be essential for higher solubilities, which is also the reason for the high NH_3 capacity of DES. The number density of hydroxy groups in ChCl/Res/Gly (0.024 mol g^{-1}) was much higher than that in hydroxy-containing ionic liquids (e.g, 0.0047 mol g^{-1} in [EtOHmim][BF_4], 0.0025 mol g^{-1} in [EtOHmim][NTf_2], 0.0037 mol g^{-1} in [EtOHmim][PF_6]). We also measured the hydrogen-bond acidity (α) of ChCl/Res/Gly using Kamlet-Taft solvatochromic method, and obtained a value of 2.87. This signifies the stronger hydrogen-bond acidity of ChCl/Res/Gly than ionic liquids (e.g., the α of [EtOHmim][NTf_2] is 1.14).[9] Thus, the stronger ability of ChCl/Res/Gly than ionic liquids to dissolve NH_3 can be ascribed to the higher abundance of hydroxy groups and higher hydrogen-bond acidity. In a very recent report, 1-butylimidazolium bis(trifluoromethylsulfonyl)imide ([Bim][NTf_2]) was reported to show NH_3 mass solubility close to ChCl/Res/Gly, which resulted from the presence of acidic $N^{\delta+}$–H groups.[4a] Nevertheless, the [NTf_2]$^-$-based ionic liquids are generally expensive for practical applications.

Promising results were obtained even when compared with the absorption of NH_3 by water, which has an NH_3 solubility of 0.30 g g^{-1} at 313 K 101 kPa.[10] Although this value is somewhat higher than the NH_3 capacity of DESs, the use of water or aqueous solutions as solvents has serious shortcomings, especially for the desorption of NH_3 and regeneration of solvents that are crucial in industrial applications.[2,3b,c,11] If hybrid DESs are employed as solvents, an NH_3 solubility comparable to that of a water system could be achieved, and the absorbent could be easily regenerated by a simple evaporation at moderate temperature while the desorbed NH_3 is obtained without contamination of the absorbent. Typically, the desorption and regeneration of DES were investigated for ChCl/Res/Gly. NH_3 was adsorbed until saturation at 313 K and 101 kPa, and then was desorbed with a vacuum pump for 5 h at 323 K. Ten absorption–desorption cycles were performed without the loss of uptake capacity (Figure 2), which suggested a recyclability very superior to aqueous solvents. Consequently, the results demonstrated that the proposed hybrid DESs not only have an advantage over conventional ionic liquids, but are also very promising even compared with aqueous absorption.

Every component in a hybrid DES is essential for its unique performance. If phenols were absent, the hybrid DES was the conventional ChCl/Gly, which already showed poor NH_3 ab-

Table 1. Solubilities of NH_3 (g g^{-1}) in different solvents at atmosphere.

Solvent	Solubility	T [K]	Ref.	Solvent	Solubility	T [K]	Ref.
[Bmim][BF_4]	0.017	298.15	[4b]	[EtOHmim][NO_3]	0.020[a]	313.15	[4b]
	0.029	313	[4c, d]		0.018[a]	333.15	[4b]
	0.0088[a]	298.4	[4h]	[EtOHmim][PF_6]	0.038[a]	313.15	[4b]
	0.0040[a]	323.6	[4h]	[EtOHmim][NTf_2]	0.036[a]	313.15	[4b]
	0.074	293	[8]		0.022[a]	333.15	[4b]
[Bmim][PF_6]	0.021[a]	298	[4h]		0.042[a]	298.15	[4b]
	0.011[a]	324.6	[4h]		0.028[a]	323.15	[4b]
[Bmim][NTf_2]	0.0053[a]	299.4	[4h]	[EtOHmim][DCA]	0.018[a]	313.15	[4b]
	0.0024[a]	323.4	[4h]		0.059	313	[4c]
	0.011	313	[4a]	[C_8mim]][BF_4]	0.0093	323.15	[4e]
[Emim][BF_4]	0.0095[a]	333.15	[4b]	[Choline][NTf_2]	0.066	313	[4c, d]
[Emim][NTf_2]	0.0065[a]	333.15	[4b]	[MTEOA][$MeOSO_3$]	0.082	313	[4c, d]
[Emim][NO_3]	0.0097[a]	333.15	[4b]	[TMGH][BF_4]	0.090	293	[8]
[Emim][Ac]	0.032[a]	298.3	[4g]	[TMGH][NTf_2]	0.050	293	[8]
	0.018[a]	324.5	[4g]	[$TMGHPO_2$][BF_4]	0.038	293	[8]
[Emim][SCN]	0.045[a]	298.1	[4g]	[DMEA][Ac]	0.10	298.1	[4g]
	0.018[a]	322.6	[4g]	[DMEA][Ac]	0.049	322.7	[4g]
[Emim][$EtOSO_3$]	0.040[a]	297.6	[4g]	[Bim][NTf_2]	0.11	313	[4a]
	0.023[a]	322.3	[4g]	[$HOOC(CH_2)_3$mim][NTf_2]	0.058	313	[4a]
[Hmim][Cl]	0.024[a]	297.8	[4h]	ChCl/Res/Gly	0.18	293.15	This work
	0.010[a]	324.3	[4h]		0.17	298.15	This work
[EtOHmim][BF_4]	0.045[a]	313.15	[4b]		0.13	313.15	This work
	0.029[a]	333.15	[4b]		0.10	323.15	This work
	0.070	313	[4c, d]		0.089	333.15	This work
[EtOHmim][SCN]	0.031[a]	313.15	[4b]				

[a] Data are taken from the figures in original publications. [b] Solubilities are given in g(NH_3) g(solvent) $^{-1}$ from the original data.

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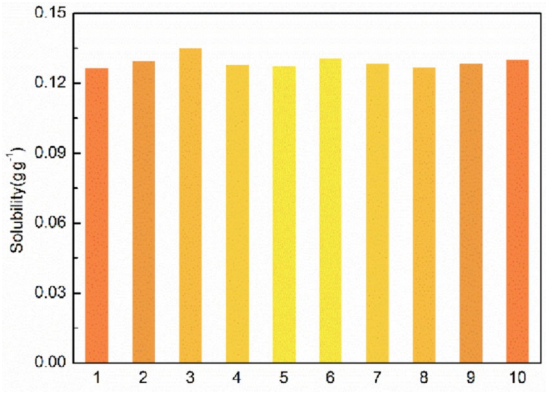


Figure 2. Solubility of NH_3 after different absorption–desorption cycles of ChCl/Res/Gly.

sorption (Figure 1a). To further explore the effect of phenols, we employed ChCl/Res/Gly in varying molar ratios of 1:1:5–1:9:5 (ChCl/Res/Gly) for NH_3 absorption at 313 K and 101 kPa. The results (Figure 3) showed that with the increase of resorcinol there was obvious increase of NH_3 capacity if the molar ratio was 1:1:5–1:7:5. However, when the ChCl/Res/Gly molar

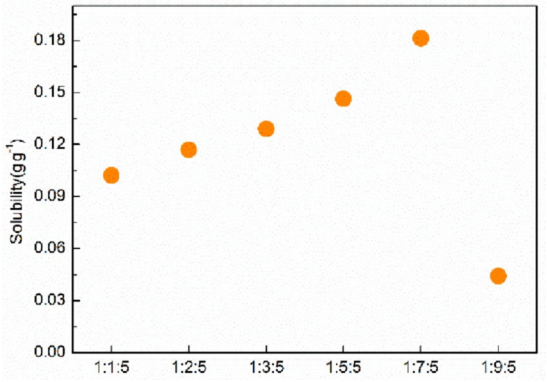


Figure 3. NH_3-absorption capacity of choline chloride/resorcinol/glycerol in varying molar ratios of 1:1:5–1:9:5 at 313 K and 101 kPa.

ratio was 1:9:5, a significant amount of white solid formed, which may indicate the breakdown of the hydrogen-bond network. Moreover, when the molar ratio further changed to 1:10:5, there was no DES generated after mixing of the three components. These results indicate the phenols played an important role in capturing NH_3 molecule, but excessive phenols would reduce the supramolecular stability of DESs during absorption and even inhibit the formation of a DES.

If polyalcohols were absent from the DES, the resultant ChCl/Phe was also poor at adsorbing NH_3, as described previously. To further investigate the effect of the polyalcohol content, ChCl/Phe/Gly with molar ratios of 1:3:1, 1:3:3, and 1:3:5 were applied to NH_3 absorption at 313 K and 101 kPa. The experiments showed that the NH_3 absorption capacity of the DES with ChCl/Phe/Gly molar ratios of 1:3:3 and 1:3:5 were 0.096 g g^{-1} and 0.095 g g^{-1}, respectively, whereas many white solids were generated and the capacity was 0.071 g g^{-1} if the molar ratio was 1:3:1. If DESs with ChCl/Res/Gly molar ratios of 1:3:1 and 1:3:3 were employed for NH_3 absorption at 313 K and 101 kPa, there was a tremendous amount of white solids formed and the mixture could not be stirred. If the molar ratio of ChCl/Res/Gly was 1:3:5, the solvent remained clear and transparent after NH_3 saturation. These results indicated that glycerol was necessary to prevent the breakdown of the supramolecular structure of the DESs to ensure the maximum absorption, but excess glycerol also decreased the NH_3 capacity. The minimum amount of glycerol required to prevent the formation of solid was different between phenol and resorcinol. To shed light on this phenomenon, other analogous phenols including catechol and phloroglucinol were used to prepare DESs to further investigate the NH_3 absorption. For catechol, which is an isomer of resorcinol, if a choline chloride/catechol/glycerol molar ratio of 1:3:5 was used the NH_3 uptake capacity was 0.12 g g^{-1} capacity and a clear solution was obtained. If the molar ratio of choline chloride/catechol/glycerol was 1:3:3, solids formed and the solution was murky. These results were very similar to those observed for resorcinol with the same molar ratio. Thus, with the increase of the amount of hydroxy groups, the choline chloride/phenols/glycerol system formed more solids when NH_3 was dissolved, and the minimum amount of glycerol required to prevent the formation of solids was higher. As expected, although choline chloride/phloroglucinol/glycerol in a molar ratio 1:3:5 was liquid, there was a large amount of white solid formed as soon as NH_3 was bubbled into the solution. The absorption process subsequently stopped and a low NH_3 solubility was obtained. However, the DES with a choline chloride/hydroquinone/glycerol molar ratio of 1:3:5 and 1:3:9 was solid under ambient conditions. This unexpected result suggested the significance of the location of the hydroxy group in the phenols in the formation of a DES. Hydroquinone had the highest melting point among the three isomers (resorcinol (T_m = 384 K), catechol (T_m = 378 K) and hydroquinone (T_m = 445 K), which may account for the failed synthesis of the choline chloride/hydroquinone/glycerol DES.

Choline chloride was also essential. When choline chloride was absent, the resultant solvent was volatile, showing a remarkable loss of weight during desorption, whereas the weight change of the ChCl/Phe/Gly and ChCl/Res/Gly DESs were negligible under the same conditions even after ten absorption–desorption cycles.

Molecular dynamics simulations

To shed light on the microscopic structural information of the hybrid DESs and their interaction with the NH_3 molecule, molecular dynamics (MD) simulation was performed for ChCl/Res/Gly before and after NH_3 absorption. Site-to-site radial distribution function (RDF) was employed to characterize the microstructural information. The atomic numbers of ChCl/Res/Gly and NH_3 are defined in Figure 4a. Figure 4b–f shows the RDFs of pure ChCl/Res/Gly before the NH_3 absorption at 313 K and 101 kPa. There were abundant hydrogen bonds among choline chloride, resorcinol, and glycerol, which indicated that a com-

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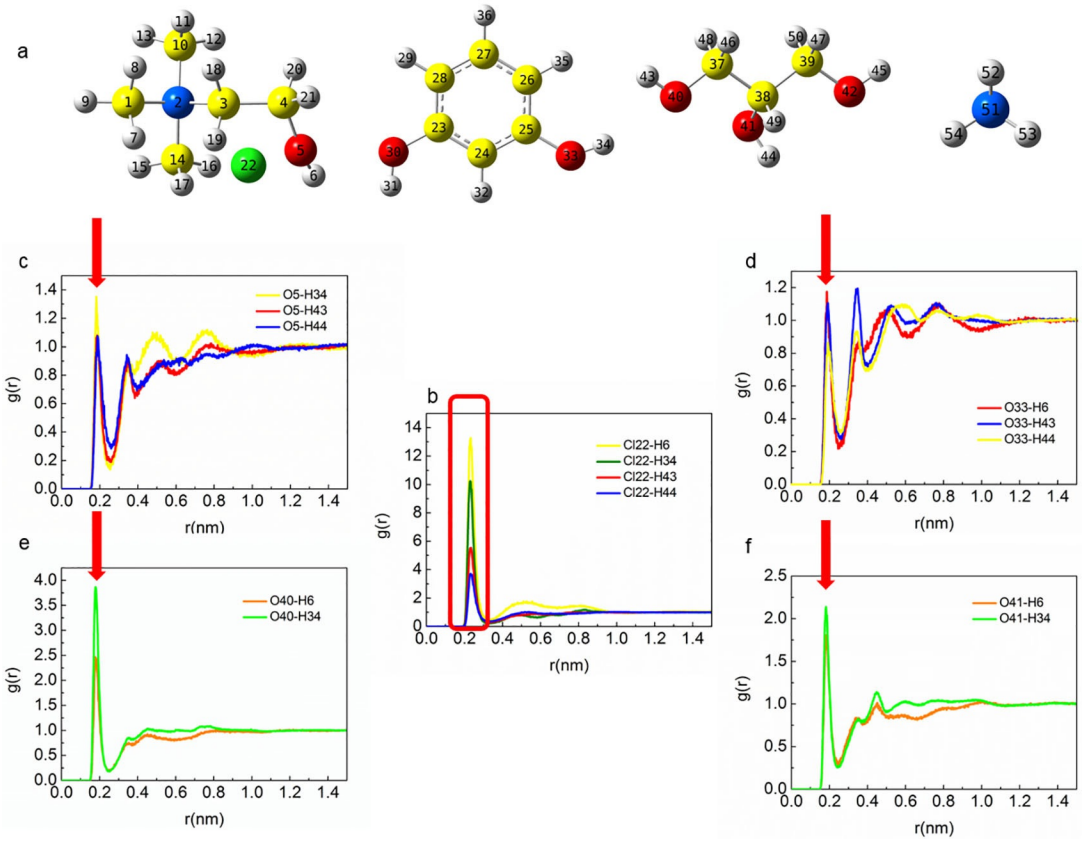


Figure 4. a) Atom numbers of choline chloride, resorcinol, glycerol, and NH_3. b–f) The radial distribution function (RDF) of pure ChCl/Res/Gly before the NH_3 absorption at 313 K and 101 kPa, in which the first peaks are highlighted by red arrows and a rectangle.

plex hydrogen-bond network was present in the system. Firstly, as presented in Figure 4b, a very high peak was observed at approximately 0.18 nm in all the RDFs between the chloride anion of choline chloride (Cl22) and the hydroxy hydrogen atoms in the cholinium cation (H6), resorcinol (H34), and glycerol (H43, H44), indicating that strong hydrogen bonds existed between the chloride and all the hydroxy hydrogen atoms in the DES (both phenolic hydroxy groups and alcoholic hydroxy groups). Secondly, as shown in Figure 4b–f, the marked first peak is present in all the RDFs between the hydroxy oxygen atoms or chlorine atoms and the hydroxy hydrogen atoms. As implied by the almost completely overlapping first peaks, the O5 in choline chloride had very similar interactions with the phenolic H34 in resorcinol or the alcoholic H43/H44 in glycerol despite the much stronger H-bond donor ability of the phenolic hydroxy compared with the alcoholic hydroxy (Figure 4c). Similar but weaker peaks at 0.18 nm were present between O33 in resorcinol and the H atoms in choline chloride and glycerol (Figure 4d). Additionally, a much more intense association existed between the oxygen atoms in glycerol (either O40 or O41) and the H in the other two compounds (Figure 4e,f) than the O5-H and O33-H interactions, implying the considerable role of glycerol in constructing the hybrid DES. Overall, this complex hydrogen-bonded supramolecular network of electronegative atoms and H atoms makes the three components of ChCl/Res/Gly strongly associated with each other, which perturbs the intrinsic microstructure of the individual components to form a deep eutectic solvent.

Figures 5 and 6 show the RDFs of ChCl/Res/Gly after the NH_3 absorption at 313 K and 101 kPa, with seven NH_3 molecules in one ChCl/Res/Gly (which is in accordance with the experimental ratio), from which abundant DES–NH_3 intermolecular hydrogen bonds were discovered. Firstly, on account of the existence of a high abundance of hydroxy groups in the DES and the basicity of the N in NH_3, the DES was prone to form hydro-

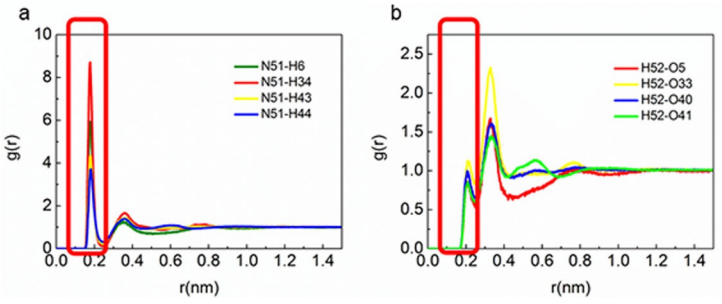


Figure 5. a) The radial distribution function (RDF) between N51 atom from NH_3 and hydroxy H from ChCl/Res/Gly. b) The RDF between the H52 atom from NH_3 and the hydroxy O from ChCl/Res/Gly, in which the first peaks are highlighted by red rectangles.

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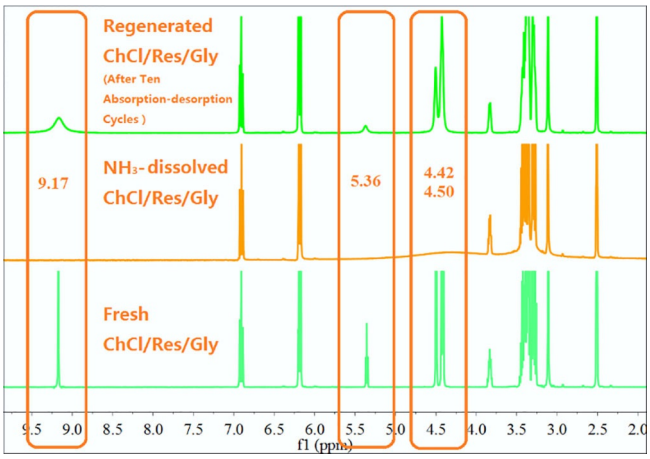


Figure 6. The radial distribution function (RDF) of ChCl/Res/Gly after the NH_3 absorption at 313 K and 101 kPa, in which the first peaks are highlighted by red arrows and a rectangle.

gen bonds between every hydroxy hydrogen in DES and the ammonia nitrogen, as confirmed by the marked first peak in all the RDFs in Figure 5a. Among them, the RDFs between N51 in NH_3 and hydroxy H34 in resorcinol gave the maximum peak at 0.18 nm (Figure 5a), followed by the N51–H6, N51–H43, and N51–H44 RDFs. Furthermore, although NH_3 is a typical basic molecule, the H atoms in NH_3 also have notable interactions with the electronegative O atoms in ChCl/Res/Gly on account of the presence of a first maximum at 0.18 nm in the H52–O RDFs (Figure 5b). In these interactions, the hydrogens in NH_3 act as hydrogen bond donors and the O and Cl in the DES act as hydrogen bond acceptor sites. Although this H_{NH_3}–X_{DES} interaction seems weaker than the N_{NH_3}–H_{DES} interaction, the fact that each NH_3 molecule has three H atoms must be taken into account in the dissolution process. The results in Figure 5 collectively manifest that there are multiple hydrogen bonds between every atom (N and H) of the NH_3 molecule and the H and O atoms of the DES, which enables strong solvation of the NH_3 molecule by the hybrid DES without any reactions.

The Cl–H and O–H RDFs from the ChCl/Res/Gly–NH_3 system are also important (Figure 6), as they show the internal site-to-site interactions of the DES. Overall, there were few variations in the position of the first peak before (Figure 4b–f) and after (Figure 6a–e) NH_3 absorption for all the RDFs, but the variation of the peak intensity was very different for the different X–H pairs. Specifically, the intensity of the first peak of the all the Cl–H RDFs was almost unchanged by the absorption, implying an undisturbed solvation shell around the Cl anion (Figures 4b, 6a). The peak intensity of the O5–H RDFs becomes notably weaker after NH_3 is dissolved, especially the O5–H34 RDF (Figure 4c, 6b) and the O33–H RDFs. Additionally, the peak intensity of the RDF between O40 (or O41) and H6 does not change, but that between O40 (or O41) and H34 becomes significantly weaker. Furthermore, the considerably lower first peaks in the O5–H34, O40–H34, and O41–H34 RDFs indicate that the interactions of the phenolic hydroxy hydrogen with the oxygen in both choline chloride and glycerol are reduced in the presence of NH_3, which is ascribed to the strong interaction of the phenolic H with the N in NH_3. Furthermore, the unchanged or less weakened first peaks in the other RDFs indicate that the overall mutual interactions among the three components of ChCl/Res/Gly were not significantly affected by NH_3 and the hydrogen-bonded supramolecular network of ChCl/Res/Gly was maintained (i.e., "flexible"), which accounts for the structural stability of the hybrid DESs during NH_3 absorption without solid formation.

NMR spectroscopy

^{1}H NMR spectroscopy was performed for fresh ChCl/Res/Gly, NH_3-dissolved ChCl/Res/Gly, and the regenerated ChCl/Res/Gly after ten absorption–desorption cycles (Figure 7, Figures S2–S4 in the Supporting Information). After dissolving NH_3, the peaks at $\delta = 9.17$, 5.36, 4.42, and 4.50 ppm in the fresh ChCl/Res/Gly disappeared, whereas a very broad flat peak appeared in the range of $\delta = 5.49$–3.77 ppm. Then, when the dissolved NH_3 was removed, even after ten absorption–desorption cycles, the spectrum of the regenerated ChCl/Res/Gly returned to be almost the same as that of fresh ChCl/Res/Gly, except that the peaks at $\delta = 9.17$, 5.36, 4.42, and 4.50 ppm were much wider than in the fresh one; the integrals of these peaks did not change. The peak at $\delta = 9.17$ ppm was assigned to the hydroxy H in resorcinol, and the peaks at $\delta = 5.36$, 4.42, and 4.50 ppm were assigned to the hydroxy H in choline chloride and glycerol. Thus, these results confirm two points. First, the absorption of NH_3 did not decompose the ChCl/Res/Gly, and the dissolved NH_3 had strong hydrogen-bond interactions with the hydroxy H atoms in ChCl/Res/Gly. Second, consistent with the regeneration experiments, the absorption of NH_3 in ChCl/Res/Gly was readily reversible and recyclable by simple evaporation. The broader hydroxy H peaks after desorption probably arise from a reconstruction of the local microstructure in the solvent.

Natural hybrid DESs for NH_3 absorption

In light of the hazardous properties of phenol and resorcinol, the feasibility of replacing them with natural compounds to produce "greener" hybrid DESs for NH_3 absorption was investigated. For all phenol substitutes, the choline chloride/phenol substitute/glycerol molar ratio was fixed to be 1:3:5 to observe whether a DES was formed and conduct the absorption. At first, 11 kinds of natural carboxylic acids and sugars with a high abundance of hydroxy groups were investigated. Some of them failed to form liquid DESs at a ratio of 1:3:5 at 313 K, including citric acid, succinic acid, D-(+)-glucose, and D-(+)-xylose. Other phenol substitutes successfully formed DESs, and the NH_3 capacities of the resultant DESs at 313 K and 101 kPa are given in Table 2. Although the performance of most of those DESs was not satisfactory, choline chloride/D-fructose/glycerol (1:3:5) was found to be an efficient solvent, which had a NH_3 uptake capacity (0.11 g g^{-1}) that was very close to ChCl/Res/Gly (0.13 g g^{-1}). This confirmed the feasibility of synthesizing an efficient hybrid DES completely from low-toxicity compounds. Interestingly, the slight difference between the structures of D-fructose, D-(+)-glucose anhydrous,

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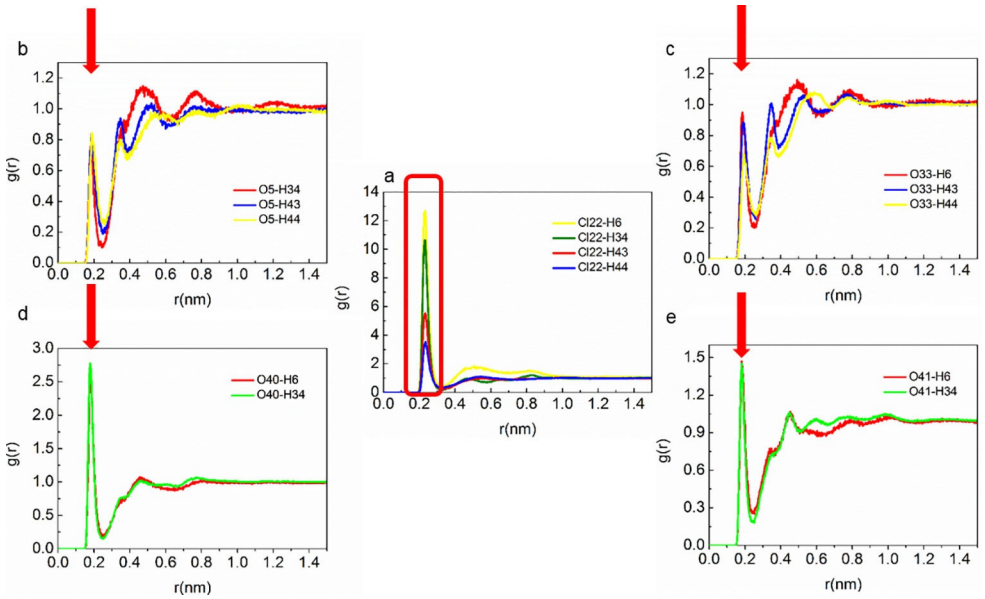


Figure 7. ^{1}H NMR spectra of fresh, NH_3-dissolved, and regenerated ChCl/Res/Gly after ten absorption–desorption cycles. The original spectrum and data are given in the Supporting Information.

Table 2. The composition and NH_3 solubility of choline chloride/phenols substitutes/glycerol (1:3:5, molar ratio) at 313 K and 101 kPa.

Composition	Solubility [g g^{-1}]
choline chloride/DL-malic acid/glycerol	0.045
choline chloride/levulinic acid/glycerol	0.055
choline chloride/oxalic acid/glycerol	0.074
choline chloride/malonic acid/glycerol	0.081
choline chloride/phenylacetic acid/glycerol	0.097
choline chloride/d-fructose/glycerol	0.11
choline chloride/resorcinol/glycerol	0.13
choline chloride/procyanidine/Glycerol	–[a]
choline chloride/succinic acid/Glycerol	–[a]
choline chloride/citric acid/Glycerol	–[a]
choline chloride/D-(+)-xylose/glycerol	–[a]
choline chloride/ D-(+)-glucose anhydrous/glycerol	–[a]

[a] Not a DES under ambient conditions.

and D-(+)-xylose led to distinct properties, implying that the formation of a low-melting point supramolecular structure after mixing individual components is a very delicate process. The discrepancy of the melting point between D-fructose (T_m = 376–378 K, decomposition), D-(+)-glucose (T_m = 419 K), and D-(+)-xylose (T_m = 418–421 K) may account for this difference.

An attempt was also made to form a DES with procyanidine, another natural flavonoid compound. However, probably owing to the high melting point and large molecular size of procyanidine, a procyanidine-based DES was only obtained with a choline chloride/procyanidine/glycerol molar ratio of 5:1:5, and the NH_3 solubility was only 0.044 g g^{-1} (Table 1). It is supposed that the increase of Cl^- density in DES is beneficial for breaking the intrinsic structure of procyanidine and forming the DES through more Cl···H–O hydrogen bonds, but this may also limit the absorption performance owing to reduced availability of phenolic hydroxy groups for the NH_3 molecule. Although many other natural flavonoids, such as anthocyanins and genistein, may also form DESs with choline chloride and glycerol, they are very expensive and these experiments were not performed.

Selectivity of NH_3 to CO_2

In addition to the absorption of NH_3, we investigated the uptake of CO_2 by the hybrid DESs to assess their potential for the selective separation of NH_3 from CO_2 because this is one of the most challenging separation tasks involving NH_3. In contrast to almost all the reported work on the uptake of CO_2 by novel materials,[7,12] which aim at enhancing the CO_2 capacity, a poor CO_2 uptake in the solvent is preferred to give a large separation selectivity in this case. As shown in Figure S5 and Figure 8, the absorption capacity of CO_2 in ChCl/Res/Gly was

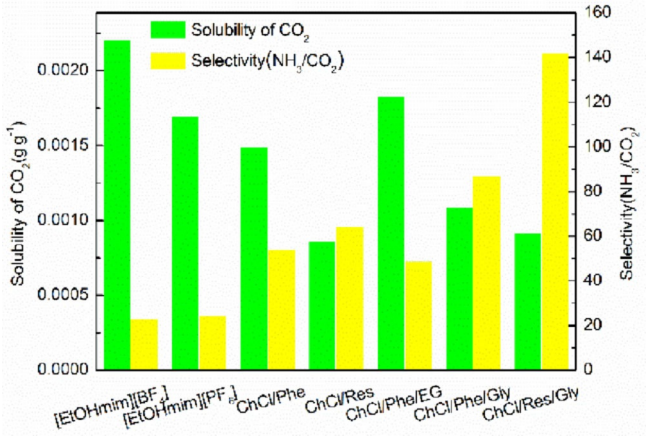


Figure 8. CO_2 absorption capacity and NH_3/CO_2 selectivity of the DESs and ionic liquids at 313 K and 101 kPa.

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0.00091 $g\,g^{-1}$ at 313 K and 101 kPa. As we have mentioned previously, the absorption capacity of NH_3 in ChCl/Res/Gly was 0.13 $g\,g^{-1}$ under the same conditions. Thus, the ideal selectivity of NH_3 to CO_2 reached up to 142. Actually, the solubility of CO_2 in ChCl/Res/Gly was the lowest value among all the DESs that have been reported to uptake CO_2,[7,12] which is likely caused by the strong hydrogen-bonded supramolecular network of the hybrid DESs, which goes against the solvation of nonpolar CO_2.

It is interesting that the binary DES ChCl/Res showed almost the same CO_2 solubility (0.00086 $g\,g^{-1}$) as ChCl/Res/Gly but lower NH_3/CO_2 selectivity (64), signifying that the addition of glycerol to ChCl/Res did not notably improve the CO_2 dissolution but contributed to a much higher NH_3 uptake. When the resorcinol in ChCl/Res/Gly was replaced by phenol, not only less NH_3 was dissolved (Figure 1 a), but a higher CO_2 solubility was also obtained (0.0011 $g\,g^{-1}$), which led to a lower NH_3/CO_2 selectivity (87). Similarly, the CO_2 solubility increased from 0.00086 $g\,g^{-1}$ to 0.0015 $g\,g^{-1}$ when the resorcinol in ChCl/Res was replaced with phenol, whereas the NH_3/CO_2 selectivity decrease from 64 to 54. When the glycerol in ChCl/Phe/Gly was replaced by ethylene glycol, increased CO_2 solubility (0.0018 $g\,g^{-1}$) and decreased NH_3/CO_2 selectivity (49) were obtained. However, the NH_3/CO_2 selectivity was still much higher than those of the hydroxy-containing ionic liquids [EtOHmim][BF_4] and [EtOHmim][PF_6]. Besides lower NH_3 solubilities (Figure 1 a), more CO_2 could be dissolved by [EtOHmim][BF_4] and [EtOHmim][PF_6] (0.0022 $g\,g^{-1}$ and 0.0017 $g\,g^{-1}$, respectively). Thus, the NH_3/CO_2 selectivities were approximately 20. These results not only show the great potential of the synthesized DESs for the separation of NH_3 and CO_2, but are also for other processes in which a low CO_2 uptake capacity is preferred.

Conclusions

In this work, a new method to capture NH_3 from gas phase with high efficiency has been proposed by using hybrid DESs as absorbents. The essence of this method is the construction of hybrid DESs with a flexible hydrogen-bonded supramolecular network, which enables the strong physical reversible solvation of NH_3 and removes the need for a trade-off between the strength of the NH_3–DES interactions and the stability of the supramolecular structure. Consequently, both exceptional NH_3 mass capacity and excellent desorption–regeneration performance were achieved by this method, along with superb NH_3/CO_2 selectivity and environmental merit. Furthermore, taking into account their easy synthesis, relativity low cost, and good biocompatibility, the hybrid DESs have great potential as efficient and green alternatives to aqueous solvents and ionic liquids in practical NH_3 capture and separation processes. Additionally, considering that the fragility of the supramolecular structure is an intrinsic character of ordinary DESs, this work is also important for the application of DESs in more processes.

Experimental Section

Chemicals and materials

NH_3 (gas, 99.999%) and CO_2 (gas, 99.00%) were purchased from Jingong Special Gas Co., Ltd. Choline chloride was purchased from J&K Scientific Ltd. Resorcinol was purchased from Sigma–Aldrich. Glycerol (99%), phenol (99%), ethylene glycol, D-(+)-glucose anhydrous (AR), D-fluctose (BR), D-(+)-xylose (BR), DL-malic acid, succinic acid (AR), and dichloromethane were purchased from Sinopharm Chemical Reagent Co. Ltd. Hydroquinone (AR), catechol (AR, 99.0%), and oxalic acid (AR, 98%) were purchased from Aladdin Industrial Corporation. Phloroglucinol anhydrous (99.0%) was purchased from TCI (Shanghai) Developmrnt Co., Ltd. Proanthocyanidins was purchased from Shanghai Macklin Biochemical Co., Ltd. 1-ethoxyl-3-methylimidazolium-tetrafluoroborate ([EtOHmim][BF_4]), 1-ethoxyl-3-methylimidazolium-hexafluorophosphate ([EtOHmim][PF_6]), 1-ethoxyl-3-methylimidazolium-bis(trifluoromethylsulfonyl)imide ([EtOHmim][NTf_2]) were purchased from Lanzhou Greenchem ILS, LICP, CAS. *N*,*N*-diethyl-4-nitroaniline was purchased from Oakwood Products, Inc. Reichardt's dye was purchased from Chemika. The ionic liquids were dried under high vacuum at 313 K for 5 h before use. The other reagents were used as received without further purification.

Synthesis of DESs

DESs were prepared by melting choline chloride with the corresponding phenols or alcohols at a suitable temperature lower than 353 K and dried with a vacuum pump at 353 K for at least 5 h. Typically, ChCl/Res/Gly was compounded with choline chloride (1.50 g), resorcinol (3.55 g), and glycerol (4.95 g) at 353 K. The water content of the DESs were measured by a Karl- Fischer Moisture Titrator to be less than 1%, similar to the water content of ionic liquids.

Absorption apparatus and procedures

The NH_3 solubility was measured by a self-constructed apparatus (shown in Figure 9). In a typical experiment, the temperature was fixed with a thermostatic water bath, and the pressure was atmospheric pressure. The DES (about 10.0 g) was placed into a 25 mL flask. Then NH_3 was charged into the flask slowly through valve V1 and valve V2, and the gas in the flask was passed out through a hose into the aqueous hydrochloric acid solution. The mass of the added DES was predetermined by an electronic balance with an uncertainty of ±0.0001 g. When the gas entered the flask and was absorbed by the DES, the total mass increased gradually. There

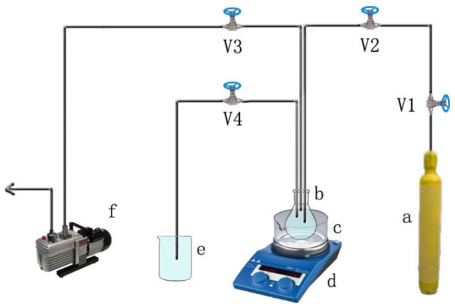


Figure 9. Apparatus schematic for NH_3 absorption. a) Ammonia gas cylinder; b) reaction flask; c) water bath; d) magnetic stirrer; e) beaker with aqueous hydrochloric acid solution. f) vacuum pump; V1–V4) adjustable valves.

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was a magnetic stirrer in the flask, which could accelerate the absorption equilibrium. The total mass was recorded every hour during the absorption process. When the total mass became constant, the equilibrium was considered to have been reached and the solubility of the DES was calculated. A desorption process was performed in the flask at 323 K for 5 h with a vacuum pump. For the DESs that showed solid formation during the absorption, the uncertainty of the NH_3 solubility was $\pm 0.004\ g\,g^{-1}$. For the other DESs and the ionic liquids, the uncertainty of NH_3 solubility was $\pm 0.001\ g\,g^{-1}$. The experimental apparatus (Figure S1) used during the CO_2 solubility measurements is based on an isochoric saturation technique. The apparatus was equipped with K-type thermocouples (accuracy ± 0.15 K) and with a pressure transducer for measuring pressures close to atmospheric pressure (Druck RPT 350, 3.5–350 kPa, accuracy $\pm 0.01\%$ full scale). The details of the absorption procedures are provided in the Supporting Information. The computation of the CO_2 solubility was performed according to the method reported in the literature.[13]

Computational method

MD simulations were run with the GROMACS 4.5.4,[14] software package. Partial atomic charges were derived from the optimized geometry using the restrained electrostatic potential (RESP) method with the R.E.D.-III.4 package.[15] To control the pressure and temperature in the NPT ensemble, a name for the isothermal–isobaric ensemble where the number of particles (*N*), pressure (*P*), and temperature (*T*) of the system are specified, the Berendsen barostat method with a relaxation time of 0.5 ps and the V-rescale thermostat method (313 K for all simulations) with a relaxation time of 0.1 ps were applied. The ChCl/Res/Gly simulation system was composed of 100 pairs of ChCl/Res/Gly (1:3:5 molar ratio). The ChCl/Res/Gly-NH_3 simulation system was composed of 100 pairs of ChCl/Res/Gly–NH_3 (1:3:5:7 molar ratio). An AMBER force field was adopted in this paper. The system was equilibrated for 2 ns in the NPT ensemble, and the last 1 ns was used to collect data for the radial distribution function and energy analysis. The forcefield parameterization used in this work was validated through the comparison of the predicted physical property, density, with the experimental value. The density of ChCl/Res/Gly obtained experimentally was approximately 1.21 $g\,cm^{-3}$ at 313 K 101 kPa, whereas the density obtained from MD was 1.24 $g\,cm^{-3}$. Likewise, the experimental density of ChCl/Res/Gly-NH_3 was approximately 1.14 $g\,cm^{-3}$ at 313 K 101 kPa, whereas the density obtained from MD was 1.20 $g\,cm^{-3}$. The deviation between the data was less than 7%, which is acceptable and shows the suitable performance of the used parameterization for describing the microscopic information. A complete listing of all the relevant force field parameters and partial charges is provided in Table S1.

Kamlet–Taft solvatochromic experiment

A Kamlet–Taft solvatochromic experiment was employed to obtain the hydrogen-bond acidity (α) of DESs, which indicates the ability of a compound to form a hydrogen bond. In this article, *N*,*N*-diethyl-4-nitroaniline and Reichardt's dye were used as the solvatochromic indicators, and the experimental details have been described in Ref. [16].

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Conflict of interest

The authors declare no conflict of interest.

Keywords: absorption · ammonia · deep eutectic solvents · molecular dynamics · supramolecular chemistry

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